



Analog IC Design

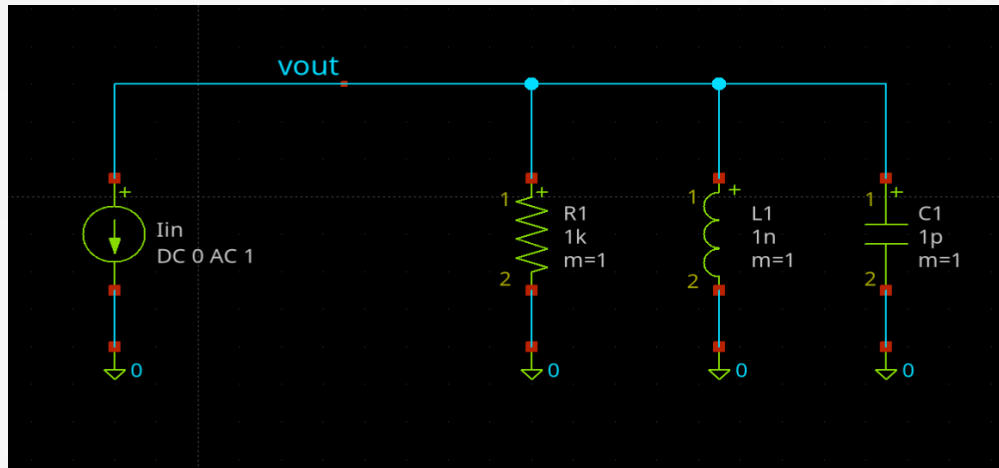
CMOS Technology

MOHAMMED NASR EL-DIN

Section 3 – Lab00

RLC circuits

Parallel RLC Circuit



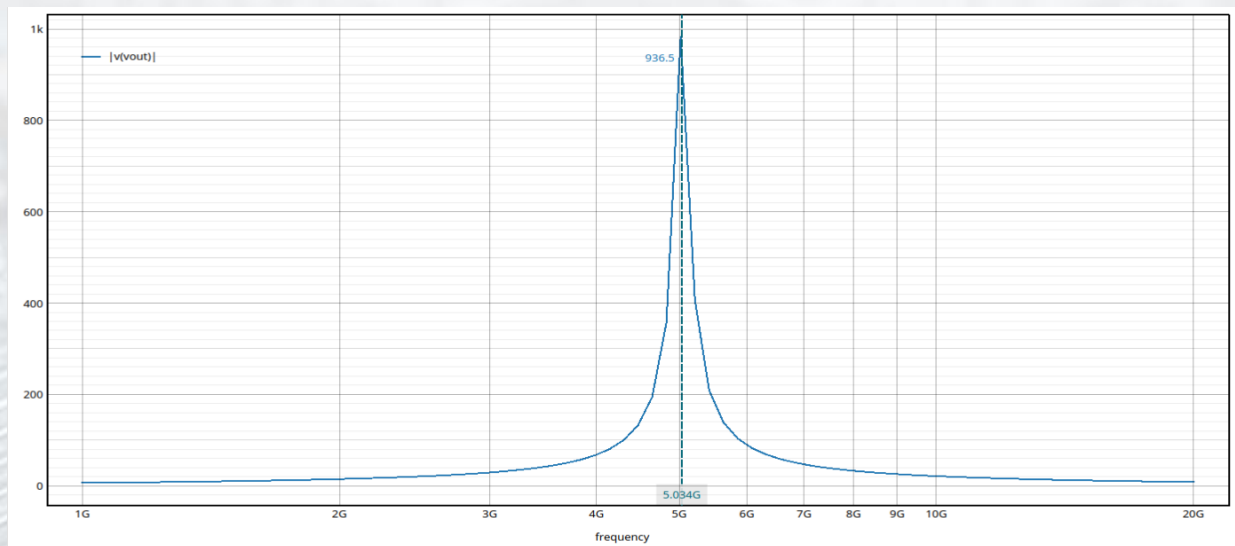
Input Impedance Expression

$$\frac{1}{Z(s)} = \frac{1}{R} + \frac{1}{sL} + sC \rightarrow Z(s) = \frac{1}{\frac{1}{R} + \frac{1}{sL} + sC}$$

$$Z(j\omega) = \frac{1}{\frac{1}{R} + \frac{1}{j\omega L} + j\omega C} \rightarrow |Z(j\omega)| = \frac{1}{\frac{1}{R} + j(\omega C - \frac{1}{\omega L})}, \omega = 2\pi f$$

AC Analysis

Applying current source with AC amplitude = 1 $\rightarrow Z_{in} = V_{out}$



Zin vs frequency

When $(\omega C - \frac{1}{\omega L}) = 0 \rightarrow$ where the impedance is pure resistive: $\omega = \frac{1}{\sqrt{LC}}$

When $(\omega C - \frac{1}{\omega L}) < 0 \rightarrow$ where the impedance is **predominantly inductive**: $\omega < \frac{1}{\sqrt{LC}}$

When $(\omega C - \frac{1}{\omega L}) > 0 \rightarrow$ where the impedance is **predominantly capacitive**: $\omega > \frac{1}{\sqrt{LC}}$

simulation outputs:

$$\begin{aligned}f_{res} &\approx 5.018 \text{ GHz} \\Z_{max} &\approx 983.4 \Omega \\BW &\approx 166 \text{ MHz} \\Q &\approx 30.1\end{aligned}$$

```
No. of Data Rows : 391
z_max      = 9.834383e+02 at= 5.018277e+09
f_res      = 5.018277e+09 with= 9.834383e+02
z_3db     = 6.953959e+02
f_low      = 4.950658e+09
f_high     = 5.117084e+09
f_res      = 5.018277e+09
z_max      = 9.834383e+02
sim_bw     = 1.664260e+08
sim_q      = 3.015320e+01
binary raw file "rlc_para_ckt.raw"
ngspice 1 -> █
```

Resonance Frequency

The resonance frequency at which Z is pure resistive:

$$\left(\omega C - \frac{1}{\omega L}\right) = 0 \rightarrow \omega_o = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1n \times 1p}} = 2\pi f_o \rightarrow f_o = 5.03 \text{ GHz}$$

The resonance frequency from the simulation: $\approx 5.018 \text{ GHz}$ small error between the simulation and the analytical results.

Quality Factor and Bandwidth

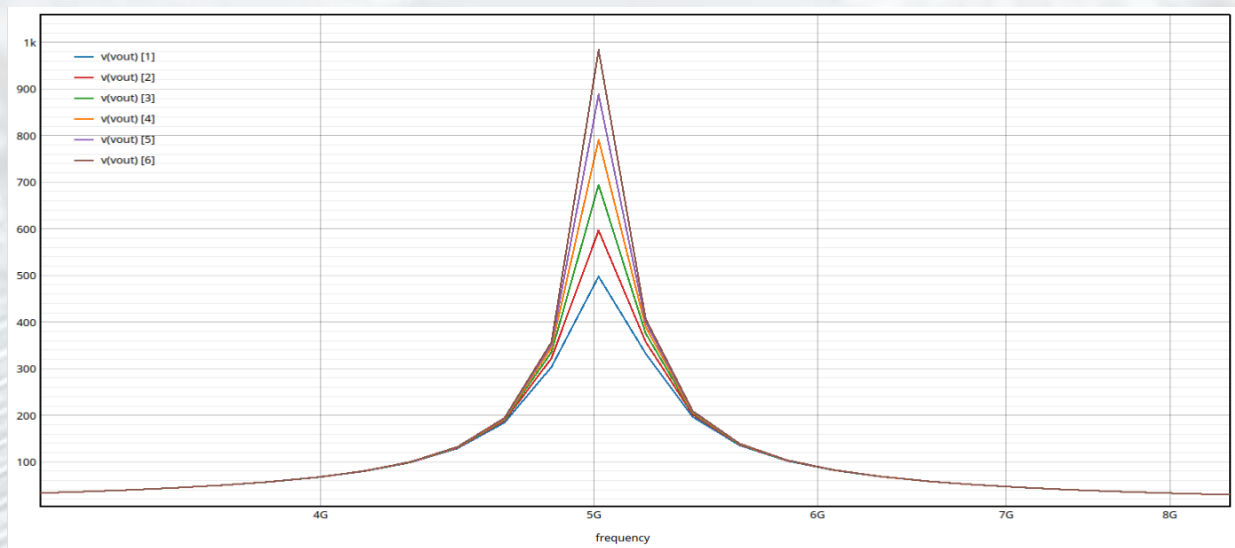
$$\text{Analytical } Q = R \sqrt{\frac{C}{L}} = 1K \sqrt{\frac{1p}{1n}} \approx 31.62$$

Q from sim $\approx 30.1 < \text{Analytical } Q$ very small error

$$\text{Analytical } BW = \frac{f_o}{Q} = \frac{5.03 \text{ GHz}}{31.62} \approx 159 \text{ MHz}$$

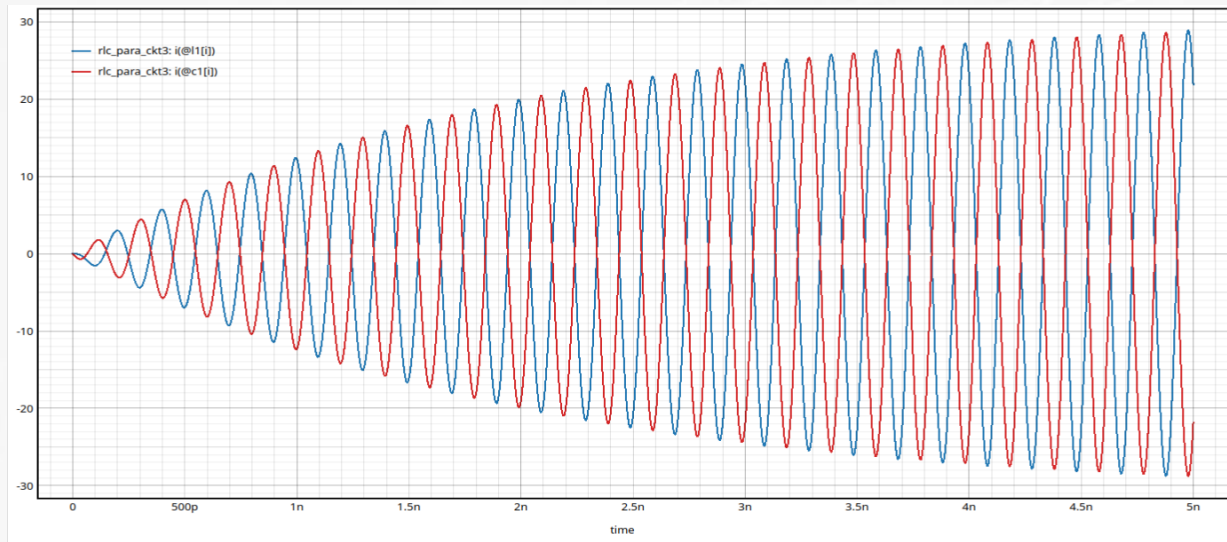
BW from sim $\approx 166 \text{ MHz} > \text{Analytical } BW$ very small error

Parametric Sweep on R



Comment: Since $Q = R \sqrt{\frac{C}{L}}$ as R increasing $\rightarrow Q$ also increasing and the resonance peak becomes more sharpness and the $BW = \frac{f_o}{Q}$ will decrease.

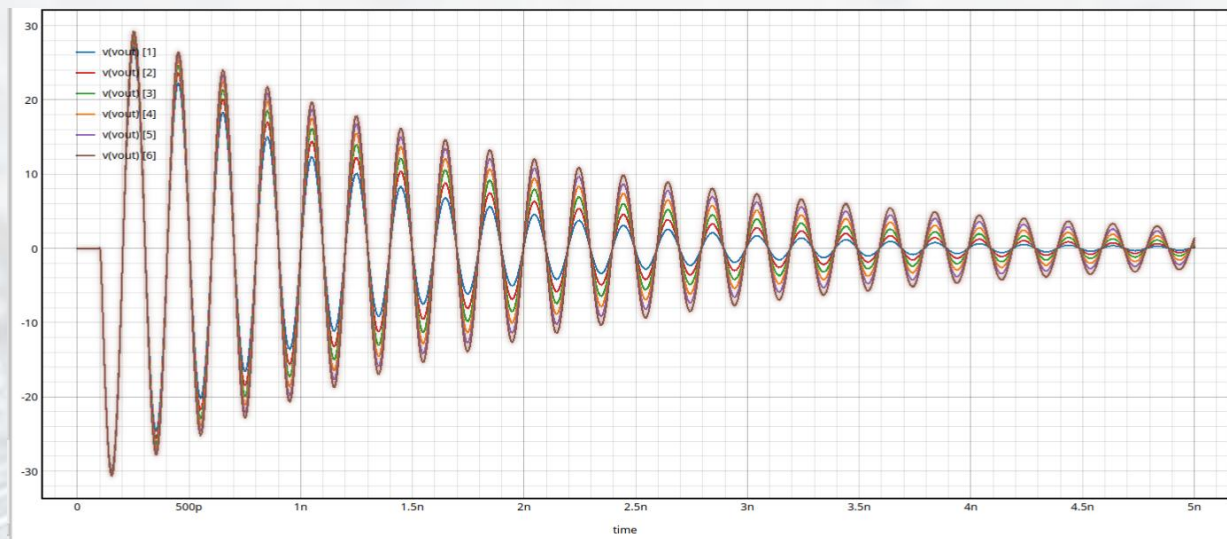
Transient response at Resonance



■ IC, ■ IL

Comment: both have the same amplitude with 180° phase shift, so both components cancel each other.

Step Response



Comment:

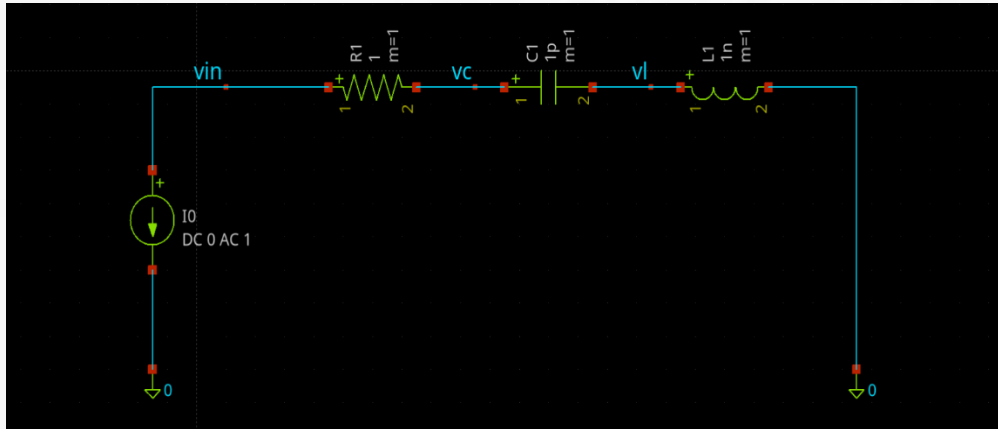
each curve damping with different damping coefficient $\alpha = \frac{1}{2RC}$.

For the curve that has the min $R = 500\Omega$, $\alpha = 10^9 \rightarrow$ largest damping coefficient $< \omega_o \rightarrow$ so all curves will show under damping response.

As $R \downarrow$, $\alpha \uparrow$ and the under damping response will be faster $\rightarrow$ Amplitude $\downarrow$

$\alpha > \omega_o \rightarrow$ over damping
 $\alpha = \omega_o \rightarrow$ critical damping
 $\alpha < \omega_o \rightarrow$ under damping

Series RLC Circuit



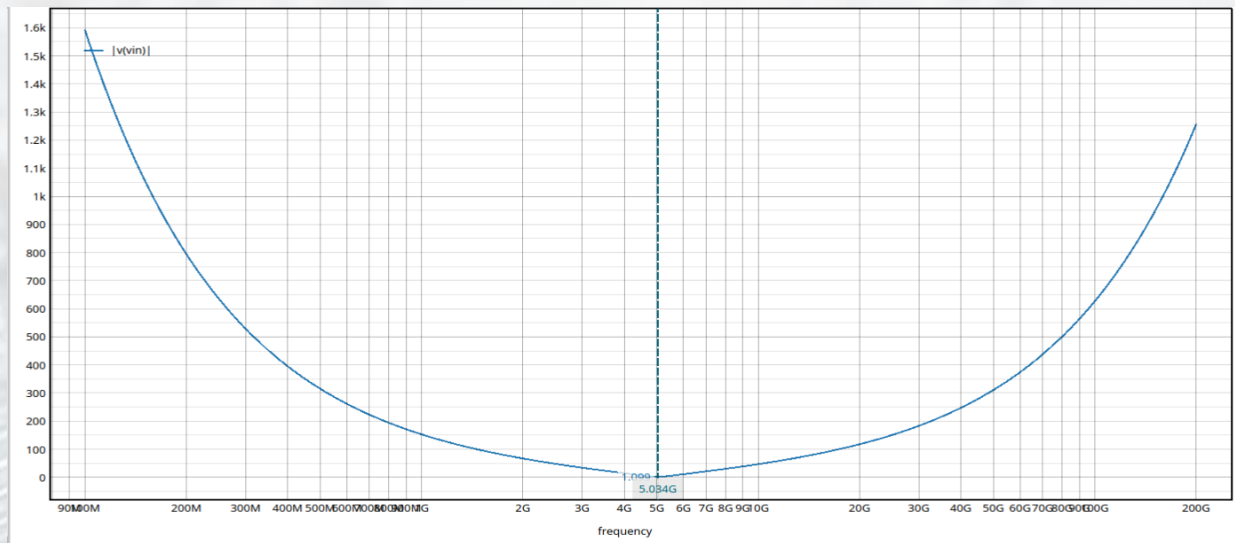
Input Impedance Expression

$$Z(s) = R + sL + \frac{1}{sC}$$

$$Z(j\omega) = R + j\omega L + \frac{1}{j\omega C} \rightarrow |Z(j\omega)| = R + j(\omega L - \frac{1}{\omega C}), \omega = 2\pi f$$

AC Analysis

Applying current source with AC amplitude = 1 $\rightarrow Z_{in} = V_{in}$



Zin vs frequency

When $(\omega L - \frac{1}{\omega C}) = 0 \rightarrow$ where the impedance is pure resistive: $\omega = \frac{1}{\sqrt{LC}}$

When $(\omega L - \frac{1}{\omega C}) < 0 \rightarrow$ where the impedance is **predominantly capacitive**: $\omega < \frac{1}{\sqrt{LC}}$

When $(\omega L - \frac{1}{\omega C}) > 0 \rightarrow$ where the impedance is **predominantly inductive**: $\omega > \frac{1}{\sqrt{LC}}$

simulation outputs:

$$f_{res} \approx 5.018 \text{ GHz}$$

$$Z_{min} \approx 1.017 \Omega$$

$$BW \approx 162 \text{ MHz}$$

$$Q \approx 30.8$$

```
No. of Data Rows : 991
z_min      = 1.017480e+00 at= 5.017999e+09
f_res      = 5.017999e+09 with= 1.017480e+00
z_3db     = 1.438934e+00
f_low      = 4.952292e+09
f_high     = 5.114822e+09
f_res      = 5.017999e+09
z_min      = 1.017480e+00
sim_bw     = 1.625300e+08
sim_q      = 3.087429e+01
binary raw file "rlc_ser_ckt.raw"
ngspice 1 -> █
```

Resonance Frequency

The resonance frequency at which Z is pure resistive:

$$\left(\omega L - \frac{1}{\omega C}\right) = 0 \rightarrow \omega_o = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1n \times 1p}} = 2\pi f_o \rightarrow f_o = 5.03 \text{ GHz}$$

The resonance frequency from the simulation: $\approx 5.018 \text{ GHz}$ small error between the simulation and the analytical results.

Quality Factor and Bandwidth

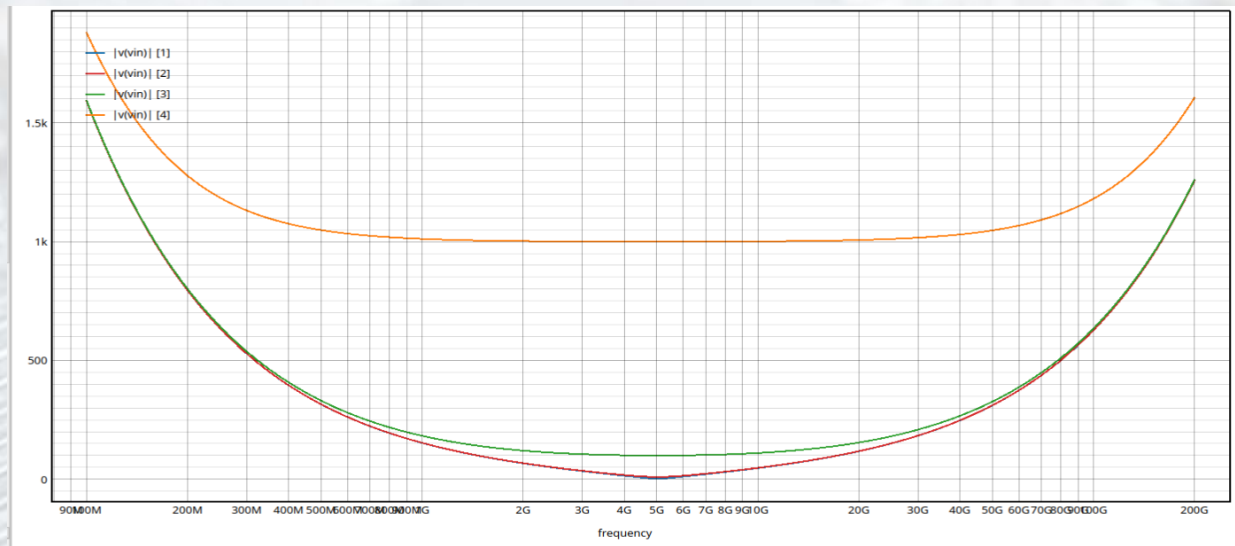
$$\text{Analytical } Q = \frac{1}{R} \sqrt{\frac{L}{C}} = 1 \sqrt{\frac{1n}{1p}} \approx 31.62$$

$$Q \text{ from sim} \approx 30.8 < \text{Analytical } Q \text{ very small error}$$

$$\text{Analytical } BW = \frac{f_o}{Q} = \frac{5.03 \text{ GHz}}{31.62} \approx 159 \text{ MHz}$$

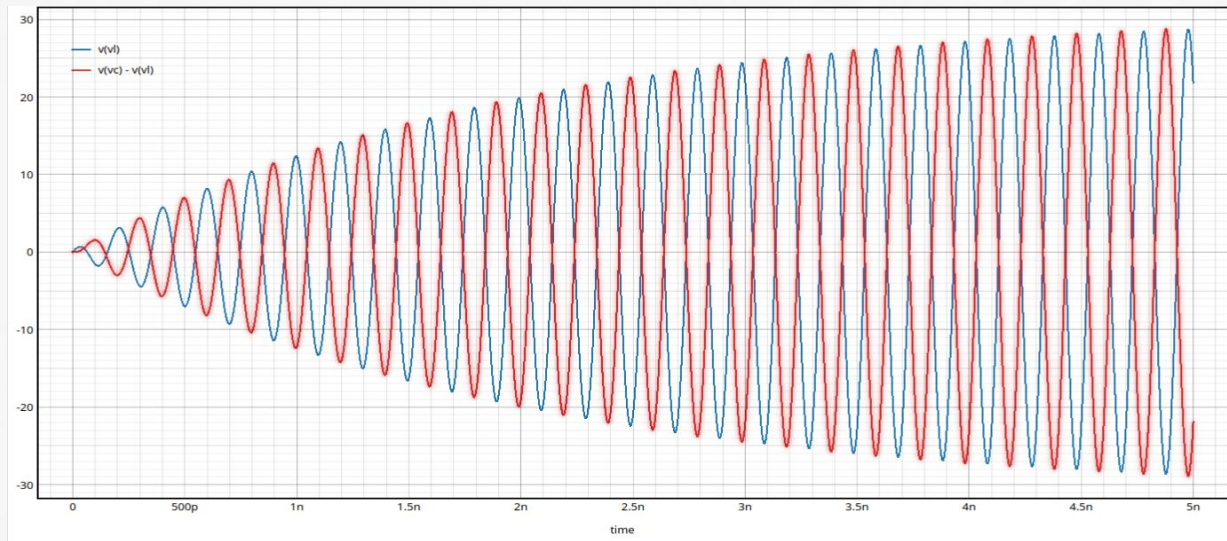
$$BW \text{ from sim} \approx 162 \text{ MHz} > \text{Analytical } BW \text{ very small error}$$

Parametric Sweep on R



Comment: Since $Q = \frac{1}{R} \sqrt{\frac{L}{C}}$ as R decreasing $\rightarrow Q$ will increase and the resonance dip becomes more sharpness and the $BW = \frac{f_o}{Q}$ will decrease.

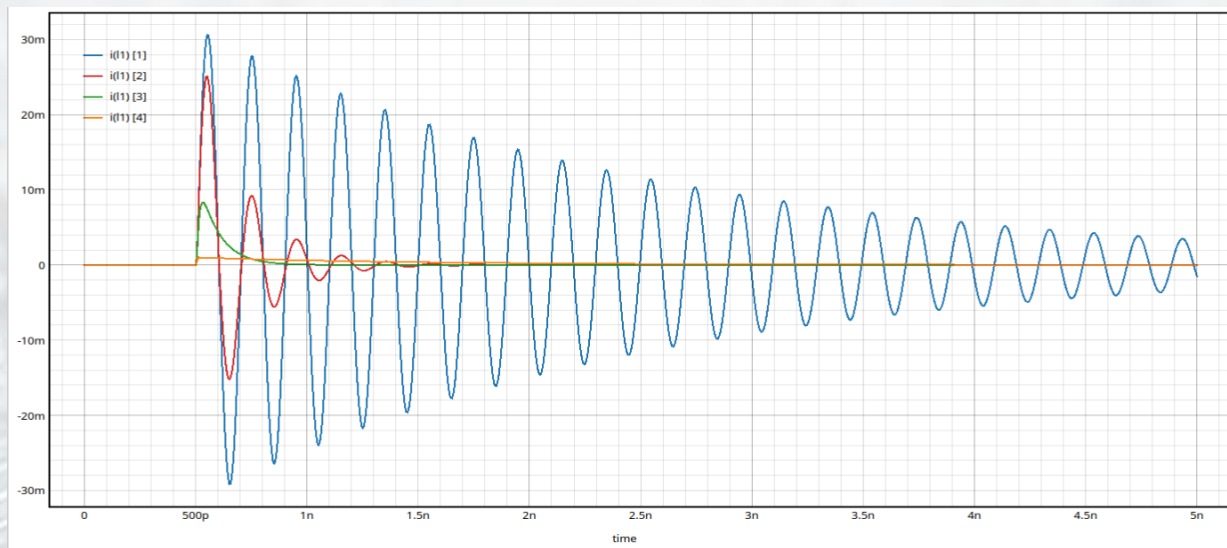
Transient response at Resonance



■ VC, ■ VL

Comment: both have the same amplitude with 180° phase shift, so both components cancel each other.

Step Response



Comment:

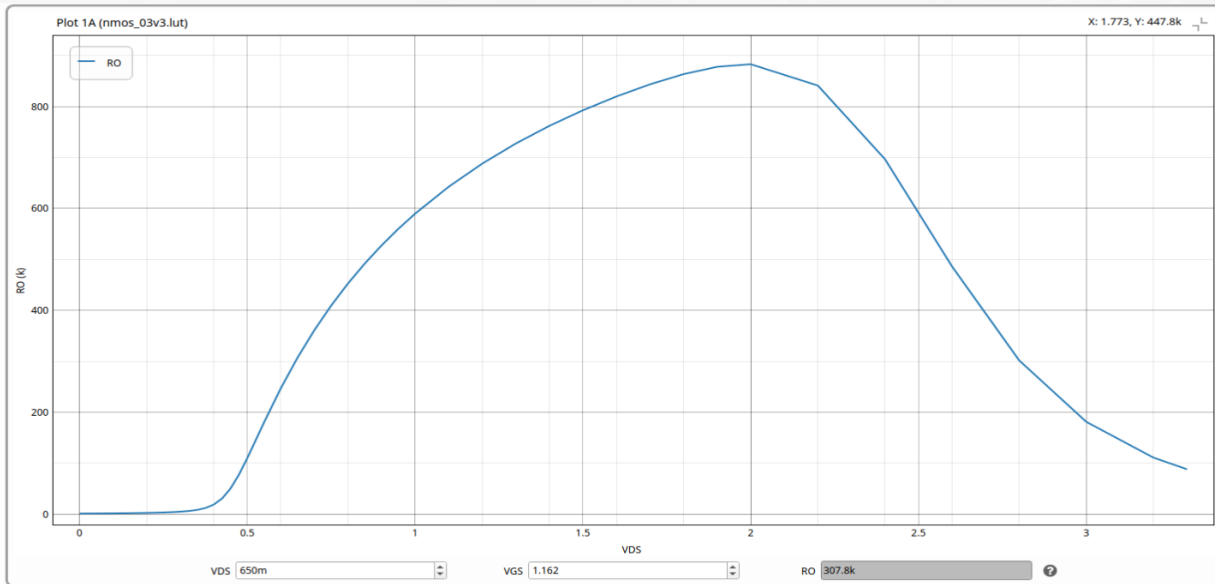
The damping coefficient $\alpha = \frac{R}{2L}$

For both curves having $R_{1,2} = 1$ and $10\Omega \rightarrow \alpha_{1,2} = 500 \times 10^6$ and $5 \times 10^9 < \omega_o \rightarrow$ under damping response.

As $R \uparrow, \alpha \uparrow$ and the under damping response will be faster $\rightarrow$ Amplitude $\downarrow$

For both curves having $R_{1,2} = 100$ and $1000\Omega \rightarrow \alpha_{1,2} = 50 \times 10^9$ and $500 \times 10^9 < \omega_o \rightarrow$ over damping response.

$\alpha > \omega_o \rightarrow$ over damping
 $\alpha = \omega_o \rightarrow$ critical damping
 $\alpha < \omega_o \rightarrow$ under damping



r_o vs V_{DS}

Comment:

- Does r_o saturate just after the transistor enters saturation similar to g_m ? Why?

No, due to channel length modulation since $r_o = \frac{1}{\lambda I_D}$, as V_{DS} highly increases the effective length will clearly decreasing so $I_D \propto \frac{1}{L}$ will increase and consequently r_o decreasing.

- Does r_o increase if the transistor is biased more into saturation?

yes, as long as $l_{eff} \approx \text{constant value}$.

- Should we operate the transistor at the edge of saturation?

No, since any small perturbation will force the transistor to leave the saturation region.

- Where do you want to operate the transistor for analog amplifier applications? Why?

In **saturation** region **away** enough from the **edge** of saturation to ensure it will remain in saturation during signal swing, and providing **highest** possible gain.